Laws of exponents

1

a a^{s+t} b a^{s-t} c a^{2s} d $2a^s$

e $a^s + a^t$ f 2^4 g $24y^4$ h $24x^7$

2

a $\frac{x^2}{4}$ b p^5 c n^7 d u^{x+1}

e p^9 f p^3 g y^7 h $24s^8t^{-2}$

i $2p^{14}$ j c^{3z+1} k $\frac{36}{m^5}$ l $21y^3$

3 $15j^{14} \div (3j^7) = 5j^7$

4

a $\frac{a^3}{b^3}$ b x^7y^7 c $16x^4$ d $\frac{x^2y^2}{25z^2}$

5 2^{16p}

6

a Multiply the exponents b Add the exponents

c Subtract the exponents d Subtract the exponents

7

a x^3 b u^{3x+3} c p^8 d p^4

e x^8 f $32y^{14}$ g $1024y^{11}$ h $108y^{10}$

i $\frac{1}{u^{10}}$ j $108m^7$ k $-135m^3$ l t^{13}

m t^{24} n u^{2x+8}

Zero and negative exponents

8

a 1 b 8 c 1 d 1

e -11 f 121

9 mn^{-4}

10

a $\frac{4}{a^2}$ b $\frac{1}{3a}$ c $\frac{6y^4}{x^3}$ d $\frac{1}{m^5}$

e $\frac{p^4}{m^5n^4}$ f $6y^4$ g $3x^7$ h $\frac{3}{x^4}$

11

a $\left(\frac{b}{a}\right)^5$ b $\frac{256}{m^{40}}$ c $\frac{m^{24}}{64}$ d $\frac{1}{125y^9}$

e $\frac{y^4}{x^2}$ f $\frac{b^{15}}{a^{15}}$ g $\frac{81}{16h^4}$ h $\frac{1}{m^{16}}$

12

a $\frac{10}{q^7}$ b $\frac{1}{8q^{10}}$ c $\frac{5z}{x^2y^7}$ d 2^{20}

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